Singapore rainfall: temporal-spatial patterns and prediction

Tasks 1 and 2: Spark exploration and probability prediction, 2017–2024

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Study objectives. Using Singapore rainfall observations from 2017–2024, we analyse temporal and spatial patterns and develop models to predict the probability that station rainfall accumulated over the next 30 minutes exceeds a given threshold.

Data and methods. The eight NEA / data.gov.sg annual datasets contain 48,538,713 source records and 48,538,627 unique valid observations from 91 station IDs. Spark performs full-data cleaning and aggregation. Annual comparisons use 21 stable stations to reduce changes caused by the monitoring network.

Main findings. Of all valid five-minute observations, 97.08% are dry. Eligible diurnal samples show the highest rain frequency at 16:00–17:00 Singapore time. Available-year monthly averages have the highest frequency in November and the lowest in July, with varying eligible years and stations. In the fixed panel, duration-normalized rainfall is lowest in 2019 and highest in 2021. Nearby stations generally have more strongly correlated daily rainfall; eight years alone do not establish long-term climate change.

Prediction results. The original algorithm comparison uses Spark MLlib Logistic, Linear SVM and external Spark XGBoost integration. Each final XGBoost model searches a complete 16-combination parameter grid using 2017–2021 training and 2022 parameter selection, then refits on all eligible 2017–2022 rows. The first half of 2023 calibrates probabilities; the second half selects probability pipelines and decision cutoffs. Final 2024 test PR-AUCs are 0.61218, 0.22511, 0.02118. The 0 mm model retains uncalibrated probabilities; 5 and 25 mm use calibration.

Year	Rain / observed 24h (mm)	Wet frequency (%)	Wet-interval intensity (mm/5min)	Mean coverage (%)
2017	6.85	2.85	0.833	94.6
2018	7.10	2.90	0.840	82.9
2019	4.81	1.97	0.841	94.7
2020	7.40	2.92	0.877	95.3
2021	7.84	3.04	0.895	97.8
2022	7.71	3.08	0.868	99.0
2023	7.52	3.16	0.824	95.2
2024	7.11	3.01	0.819	95.6

Each year in this table uses the same 21 stations, with equal station weights. Duration-normalized rainfall is neither an official annual total nor an actual daily total. Eligible monthly and hourly samples can vary.

1. Data sources, fields and cleaning

Sources and fields. Records contain observation and update timestamps, station ID/name/device ID, coordinates, reading update timestamp, rainfall amount, type, unit and date fields. There are no temperature, humidity or wind observations. API previews identify fields and annual IDs; weather findings come from the complete annual files.

Data checks. After downloading complete annual CSV files, we reconcile Spark input row counts with annual API metadata and check duplicate observations, timestamp alignment and rainfall values.

Cleaning. Spark parses timestamps in Asia/Singapore. It accepts finite, nonnegative mm readings with the expected five-minute type and valid station ID, timestamp and CSV record. Timestamps are aligned only if within two seconds of the five-minute grid; original epochs are retained. Station-time duplicates use the latest reading update, then deterministic tie-breakers. Combined annual data are checked again for cross-file duplicates.

Year	Source rows	Duplicates removed	Off-grid over 2 s	Negative readings	Stations
2017	5,256,106	86	0	0	62
2018	4,594,246	0	0	0	56
2019	5,026,862	0	0	0	54
2020	6,334,473	0	0	0	80
2021	7,044,646	0	0	0	76
2022	7,133,230	0	0	0	72
2023	6,742,890	0	0	0	74
2024	6,406,260	0	0	0	70

Cleaning results. All 86 duplicate records occur in 2017. The eight years contain no negative rainfall, timestamps more than two seconds from the five-minute grid, or cross-file duplicates. Missing intervals are not counted as zero rainfall. Additional screening of readings above 50 mm/5min identifies none.

2. Metrics and comparable samples

For a station and time period, let N be the number of valid observations, $r_i \geq 0$ the rainfall amount, $W = \sum_i \mathbf{1}(r_i > 0)$ the number of wet intervals, and E the expected calendar slots. We use:

$$\begin{aligned} \text{Coverage} &= N/E, \qquad E = 288 \times \text{calendar days}, \\ \text{Wet frequency (\%)} &= 100W/N, \\ \text{Wet-interval intensity (mm/5min)} &= \sum_i r_i/W \quad (W > 0), \\ \text{Rain / observed 24h (mm)} &= 288 \sum_i r_i/N. \end{aligned}$$

Calendar denominators handle leap years. Low coverage can reflect a station operating for only part of a year. Intensity averages wet intervals only. For one station and period, normalized rainfall equals 288 × frequency/100 × wet-interval intensity. This product generally does not hold after separate station averaging. Normalized rainfall is not an official daily or annual total and does not reconstruct missed storms.

Fixed annual panel. Of 91 station IDs across eight years, 21 satisfy all criteria: observations in every year, annual coverage of at least 80% in each year, and latitude/longitude spans no larger than 0.005°. The other 70 are excluded only from fixed-panel annual comparisons; their valid observations were not deleted during cleaning. The 21 stations receive equal weights. The next page lists the station scope for each analysis.

Monthly and diurnal samples. Figures 2 and 4 further screen sufficiently complete months or hourly strata from the 21-station pool. Monthly summaries average stations equally, then available years equally. Each eligible year-month has 17–21 stations, and each month has 6–8 available years. Requiring the same station-month to qualify in all eight years leaves only May, June, July, September, October and December, used as an additional comparison.

Spatial samples. Figure 1 uses 2024, which has 70 stations. Of these, 56 have annual coverage of at least 90% and enter map colouring; the other 14 are shown in grey. Figure 5 uses daily totals from the fixed 21 stations, requiring at least 95% daily coverage and 365 common valid days per pair. The 80%, 90% and 95% completeness thresholds are project choices.

3. Computing environment and station scope

Computing environment. Task 1 runs on an Apple Silicon Mac with 10 logical CPUs, 16 GiB RAM, macOS 26.5.2 ARM64, Python 3.12.14, Java Temurin 17.0.20.1+1 and Spark 3.5.8. Task 2 runs on gpushare with Python 3.11.10, Azul Zulu Java 17.0.20.1, Spark 3.5.8 and XGBoost 3.2.0. The gpushare container quota is 42 CPUs and about 667.68 GiB RAM; full-feature model training uses 20 CPUs and a 6 GiB default driver heap.

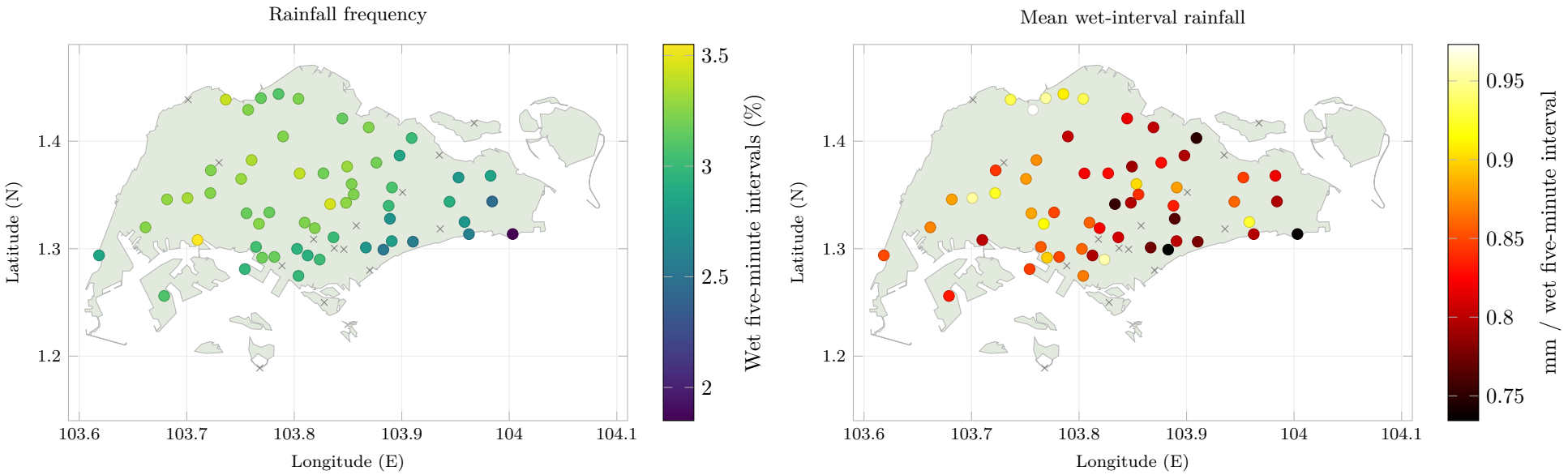
Spark configuration and responsibilities. Task 1 and the initial feature-group comparisons use single-machine `local[2]`. Full-feature model comparison and evaluation use `local[20]`, 80 shuffle partitions and default parallelism 20. XGBoost uses two workers with ten CPUs per task. Experiments run sequentially at low priority without a GPU. Spark DataFrame/SQL and windows process full-data cleaning, aggregation, labels and features; Spark ML pipelines train and evaluate models. Ordinary phases use a 6 GiB Java heap and a 12 GiB process-tree stop limit. XGBoost fitting and evaluation have a 40 GiB process-tree limit.

Analysis samples. Station scope and completeness criteria are chosen according to the comparison, as summarized below.

Analysis	Station scope	Screening rule
Figure 1: 2024 station map	56 of 70 stations that year	Annual coverage $\geq 90\%$; 14 shown in grey
Figure 2: monthly patterns	Eligible subsets of the fixed 21	17–21 stations per eligible year-month
Figure 3: annual changes	The same 21 stations	Other 70 excluded from this comparison only
Figure 4: diurnal patterns	Eligible strata of the fixed 21	Year-month-hour coverage $\geq 80\%$
Figure 5: distance and correlation	Fixed 21 stations; 210 pairs	Daily coverage $\geq 95\%$; ≥ 365 shared days
Figure 6 left: wet amounts	All 91 stations across eight years	Wet five-minute observations only
Figure 6 middle/right: exceedance and persistence	Valid 2024 observations from all 70 stations	Six complete future slots or two adjacent slots

Cleaning checks. Source row counts match API counts; 86 duplicate records were removed, and missing slots are not zero-filled. The overall dry fraction uses all 91 stations, rather than only the 21-station panel.

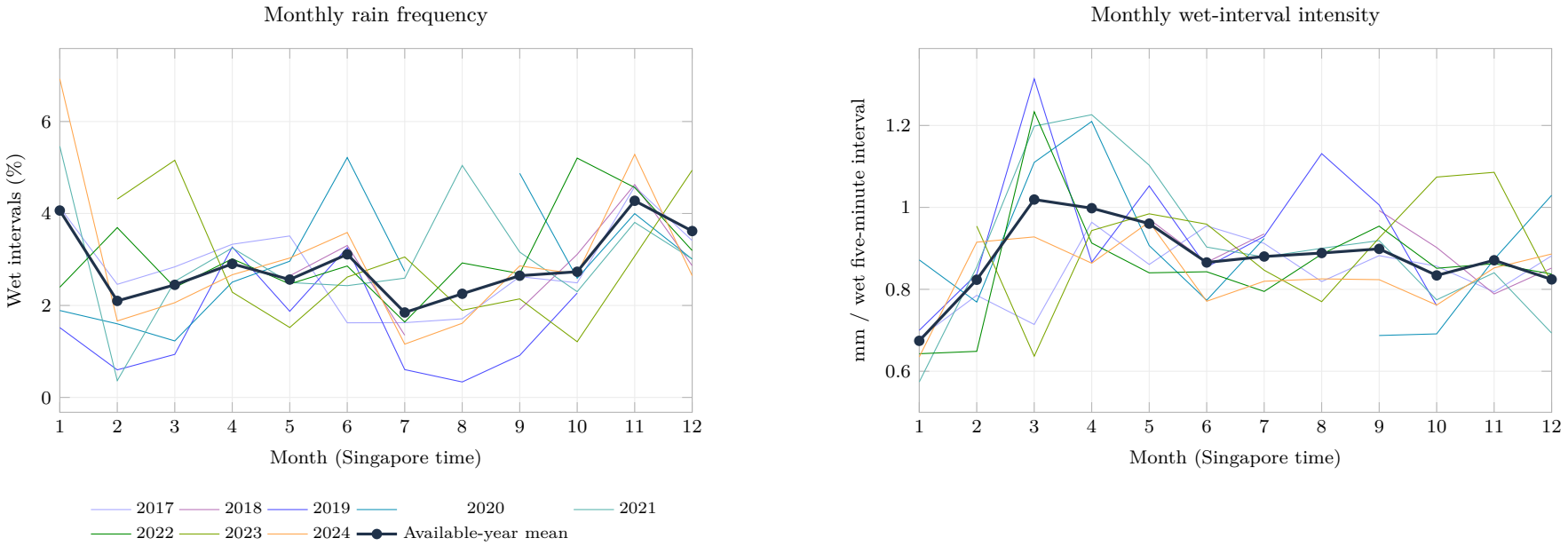
Figure 1: Station rainfall maps in 2024



In 2024, 56 of 70 stations have annual coverage of at least 90%. Coloured points show individual gauge statistics; grey crosses indicate lower coverage. S33 (Jurong Pier Road) has the highest rain frequency, 3.55%. No spatial interpolation is performed. The geoBoundaries land outline is derived from URA 2016 data under ODbL 1.0 and serves as a geographic reference.

Task 2 feature implications (stations). Candidate features include station ID and coordinates to represent station differences. Task 2 should screen windows for historical-feature and future-label completeness, rather than automatically restrict training to the 21-station annual panel.

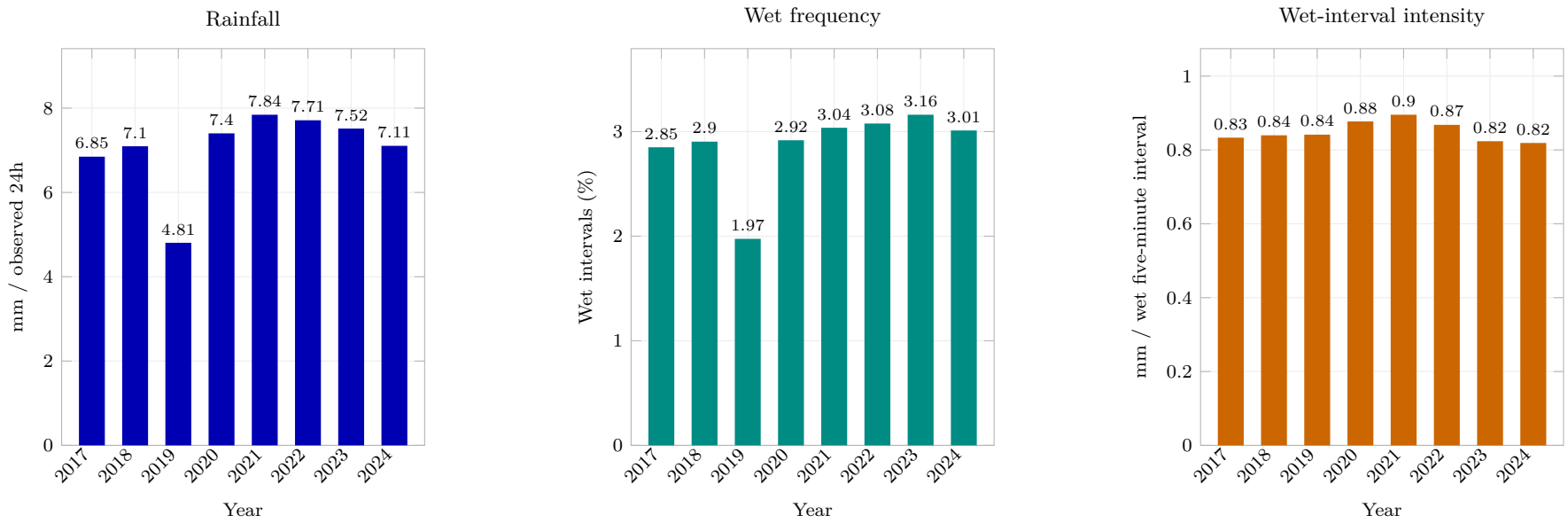
Figure 2: Monthly patterns within a year



Eligible station-months from the fixed panel require at least 80% coverage, with equal averaging over stations followed by available years. Mean monthly rain frequency is highest in November (4.28%) and lowest in July (1.85%). Mean wet-interval amounts are about 1.02 mm in March and 0.67 mm in January. Frequency and wet-interval amount show different seasonal patterns. Each month includes 6–8 years, and each eligible year-month includes 17–21 stations.

Task 2 features (month). Monthly frequency and wet-interval intensity vary differently. We encode month using sine and cosine to represent the annual cycle and combine them with hour encoding to capture seasonal variation in diurnal rainfall.

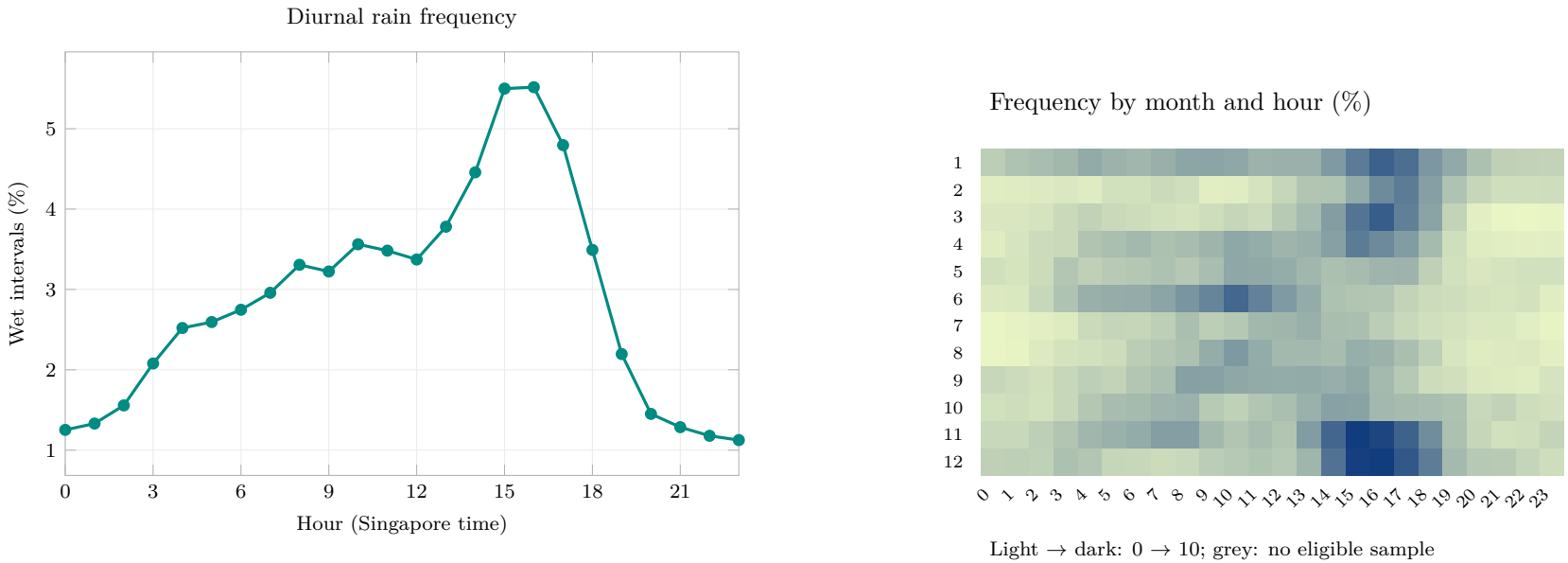
Figure 3: Annual changes in the fixed panel



Each year compares the same 21 equally weighted stations, each with annual coverage of at least 80%. Mean rainfall per observed 24 hours ranges from 4.81 mm in 2019 to 7.84 mm in 2021. Rain frequency in 2019 is 1.97%, below the other years, while its wet-interval amount of 0.841 mm is close to adjacent years. Lower rainfall frequency is thus the main component associated with its lower normalized rainfall.

Task 2 features (year). Training, validation and test are split chronologically. Models use cyclic month and hour features without year identifiers. NEA relates the dry conditions in 2019 to a strong positive Indian Ocean Dipole, providing external context for annual variation.

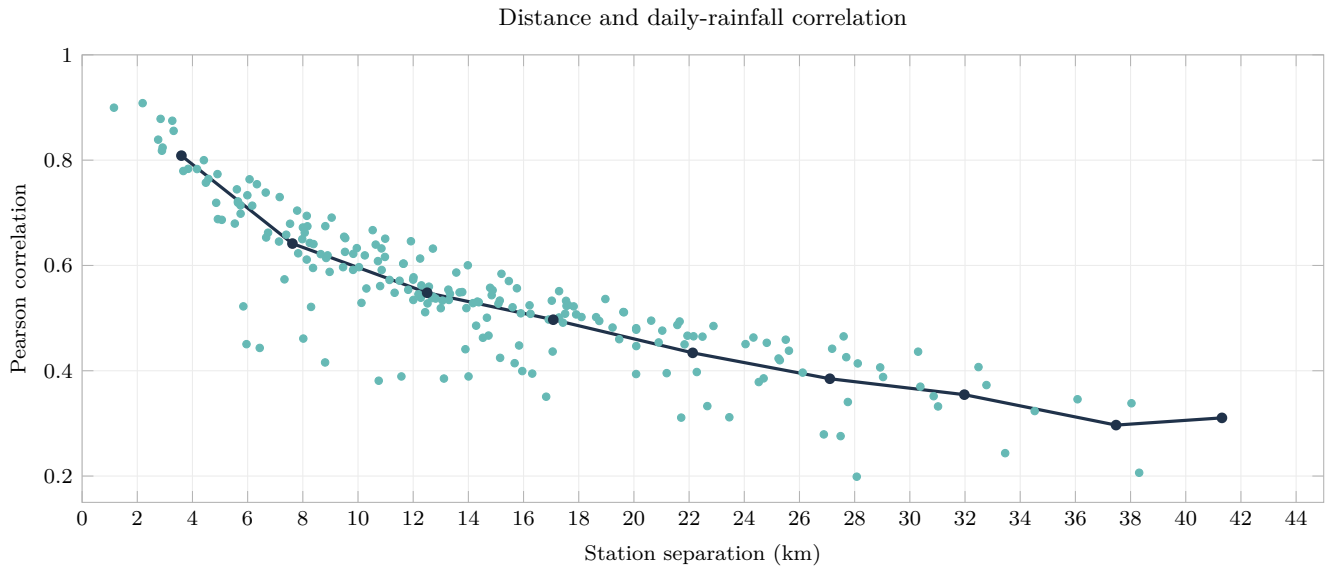
Figure 4: Diurnal variation in rainfall frequency



Rain frequency peaks at 16:00–17:00, with peak timing and strength varying by month. The afternoon peak is more pronounced in November–December, while June has elevated frequency from morning toward midday. Statistics equally weight eligible station-year-month-hour strata with at least 80% coverage. These findings motivate hour, month and their interactions as predictors.

Task 2 feature implications (hour and month). Encode Singapore local hour cyclically and consider month–hour interactions: the strength and timing of the afternoon peak vary by month. Logistic regression and linear SVM need explicit interaction features; trees can learn such combinations. Their predictive contribution must still be tested by ablation.

Figure 5: Distance and daily-rainfall correlation

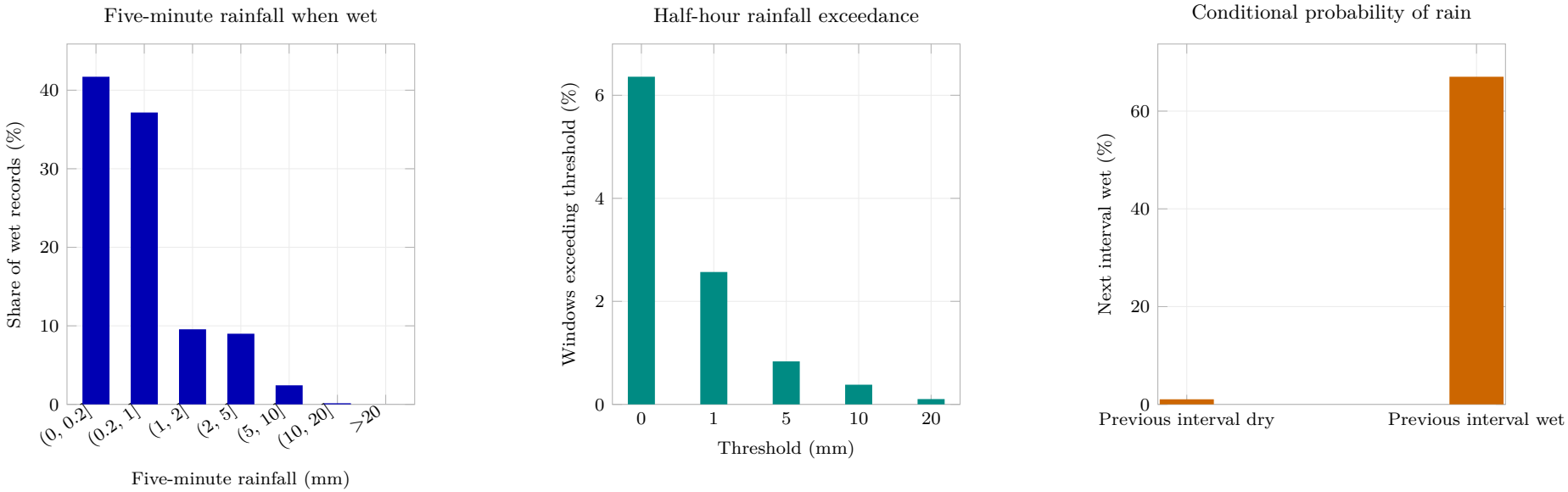


The 21 stations form 210 pairs. The mean Pearson correlation of daily rainfall is 0.809 for 17 pairs separated by at most 5 km, compared with 0.336 for 12 pairs more than 30 km apart. Mean correlations in 5 km distance groups generally decrease with separation, indicating spatial dependence.

Stations are not independent. Nearby gauges may experience the same rainfall event, and different pairs can share a station. Thus, 91 gauges are not 91 independent weather samples, and the 210 correlations are not 210 independent samples. Shared-day screening follows the criteria given earlier.

Task 2 features (neighbours). Stronger daily-rainfall correlation among nearby stations motivates mean rainfall, maximum rainfall and wet fractions from the three closest stations. Task 2 compares models before and after adding neighbour features to evaluate their contribution to half-hour prediction.

Figure 6: Rainfall distribution, half-hour exceedance and persistence



Rainfall distribution in wet intervals. Across all stations and eight years, 41.72% of wet five-minute observations fall in $(0, 0.2]$ mm and 37.16% in $(0.2, 1]$ mm. Together, 78.88% have at most 1 mm, indicating small accumulations in most wet intervals. Dry intervals account for 97.08% of all valid five-minute observations.

Exceedance probability of half-hour rainfall. There are 6,374,476 complete future six-interval windows in 2024. Subsequent 30-minute totals strictly exceed 0 mm in 6.36% of windows and 5 mm in 0.83%; exceedance becomes less frequent as the rainfall threshold rises. Totals are compared at nine decimal places to avoid floating-point errors at threshold boundaries. The horizontal axis uses threshold categories.

Rainfall persistence in consecutive intervals. For consecutive five-minute observations at the same station in 2024, the probability of rain is 1.04% following a dry interval and 67.03% following a wet interval. This substantial short-term persistence supports recent rainfall and wet-spell duration as predictors.

Task 2 feature implications (persistence and rare events). Implemented features include the latest reading, 15/30/60-minute totals, recent rainfall changes and wet-spell duration. Only records available at prediction time are used; missing readings are not dry readings. The following sections report positive counts, PR-AUC, Brier score and calibration curves separately by threshold.

4. Task 2: target and chronological splits

Prediction target. For station s, time t and rainfall threshold q, estimate the probability that rainfall in the next 30 minutes strictly exceeds q. Use $q \in \{0, 5, 25\}$ mm and one classifier per target, with one feature row and three label columns per station/time.

Threshold choices. We preset rainfall thresholds of 0, 5 and 25 mm to examine any rainfall, exceedance of a lower positive accumulation threshold, and a larger accumulation event over the next 30 minutes. These targets allow comparison of model ranking and probability quality as events become rarer. The thresholds are experimental choices, not official rainfall categories. A classifier is trained and selected separately for each threshold; the current interface supports only these three values rather than arbitrary continuous thresholds.

$$R^{30}_{s,t} = \sum_{k=1}^{6} r_{s,t+5k\,\min}, \qquad y^{(q)}_{s,t} = \mathbf{1}(R^{30}_{s,t} > q), \qquad \hat{p}_q = P(y^{(q)} = 1 \mid x_{s,t}).$$

Label construction. Treat timestamps as five-minute interval ends. Sum six valid intervals in $(t, t+30\text{min}]$ and round to nine decimal places before comparison. Exclude the current interval and reject windows missing any future interval. Equality with the threshold is negative.

Purpose	Rows	> 0 positives	> 5 positives	> 25 positives
2017–2022: all training records	34,542,285	2,074,430	279,801	19,787
2023 H1: calibration	3,325,499	251,794	30,699	1,532
2023 H2: selection	3,393,373	197,358	28,000	2,487
2024: test	6,374,476	405,430	53,193	3,517

Training and validation. The original algorithm comparison and final refits use all 34,542,285 eligible 2017–2022 rows. Grid candidates and their preprocessing use all 27,434,009 eligible 2017–2021 rows, with parameters selected on 2022. The first half of 2023 fits probability calibration and the second half selects probability pipelines and decision cutoffs; 2024 evaluates frozen models. Target windows cannot cross split boundaries. Calibration, validation and testing retain natural class proportions.

5. From Task 1 to feature engineering

Task 1 finding	Construction	Inputs
Afternoon peak varies by month	Cyclic hour/month and interactions	8 calendar features; no year ID
Station differences and spatial correlation	Station ID, coordinates, three closest neighbours	Neighbour rain means, maxima and wet fractions
Short-term persistence	Latest reading and 5/10/15/30/60-minute lags	15/30/60-minute totals, changes and wet-spell length

History and neighbours. Use only observations available by prediction time: both observation and retained update timestamps must be no later than t. Search the past 90 minutes; mark the latest reading unavailable after 30 minutes. Use coordinates known by t to identify three closest eligible neighbours and aggregate rainfall, available counts, ages and distances. Spark windows and joins process all records.

Cyclic time encoding. Let h be local hour including fractional minutes and m the month. Use the cyclic encodings below and their four pairwise products, giving eight features. Adjacent 23:00/00:00 and December/January stay close in feature space.

$$h_{\sin} = \sin(2\pi h/24), \quad h_{\cos} = \cos(2\pi h/24), \qquad m_{\sin} = \sin[2\pi(m-1)/12], \quad m_{\cos} = \cos[2\pi(m-1)/12].$$

Input processing. Impute missing values using training-period means; select missingness indicators and fit the station vocabulary on the complete training period only. Use one-hot station encoding for linear models and a categorical index for trees; exclude constant features. Logistic and SVM standardize features. Nine logistic models compare calendar/station, added own history, and added neighbour history.

Feature comparison settings. The nine ablation classifiers use L2 parameter 0.001, at most 80 iterations and tolerance 10^{-6}. 0 reach this iteration cap. Logistic and SVM in the main algorithm comparison allow 100 iterations.

Feature dimensions. The original comparison uses 45 numeric features, 27 selected missingness flags and station encoding: 162 dimensions for linear models and 73 for trees. Final grid models flag missingness for all 46 candidate numeric variables and retain 45 nonconstant numeric features plus one station category index, giving 92 dimensions. Imputation and encoding fit only the corresponding training period; differences from the original comparison cannot be attributed solely to tuning.

L2 logistic features	Test PR-AUC: 0 mm	5 mm	25 mm
Calendar/station	0.11588	0.01671	0.00139
Add own history	0.53158	0.15868	0.01482
Add neighbours	0.56134	0.17982	0.01713

Neighbour contribution. Adding neighbour history changes test PR-AUC by +0.02975, +0.02114, +0.00231 relative to own-history models, consistent with Task 1 spatial correlations.

6. Models, imbalance and calibration

Algorithm comparison setup. This section's fixed-parameter table and three policies belong to the original 27-fit algorithm comparison, not final grid models. Logistic and Linear SVM use Spark MLlib; XGBoost uses an external Spark interface. Original results are retained as baselines; final XGBoost parameters appear in Section 7.1.

Model	Parameters
L2 logistic (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
Linear SVM (MLlib)	$\lambda = 0.001$, standardization, 100 iterations, tolerance 10^{-6}
XGBoost (Spark interface)	80 trees, depth 4, learning rate 0.1, 64 bins, minimum child weight 20; $L_2 = 1$

Training policies. Compare natural distribution, balanced weighting and constrained SMOTE. Let N be the number of real training records and P the target-specific positive count. Weights are $w_+ = N/(2P)$ and $w_- = N/[2(N - P)]$. SMOTE targets 20% positives and caps their total at ten times the original count. Every policy retains all real training observations; synthetic rows enter training only.

SMOTE construction. All training positives are available as donors, processed sequentially by station to bound memory. Search up to five neighbours with identical station, missingness, discrete profile and rain-zero pattern. Interpolate 16 continuous rain features with one coefficient, preserving calendar and discrete information; copy isolated donors.

Probability calibration. Fit Platt calibration $\hat{p} = 1/[1 + \exp(-(az + b))]$ on 2023 H1, where z is the base score and $a \geq 0$. This converts SVM scores to probabilities and adjusts weighting/resampling probability scales. A group predicted near 10% should ideally contain about 10% actual positives.

7. Evaluation and validation selection

Evaluation metrics. ROC-AUC measures ranking, with random-ranking reference 0.5. PR-AUC combines precision and recall and is useful for rare positives; random-ranking precision is approximately prevalence π. Spark integrates both areas exactly. Brier score is mean squared probability error; lower is better.

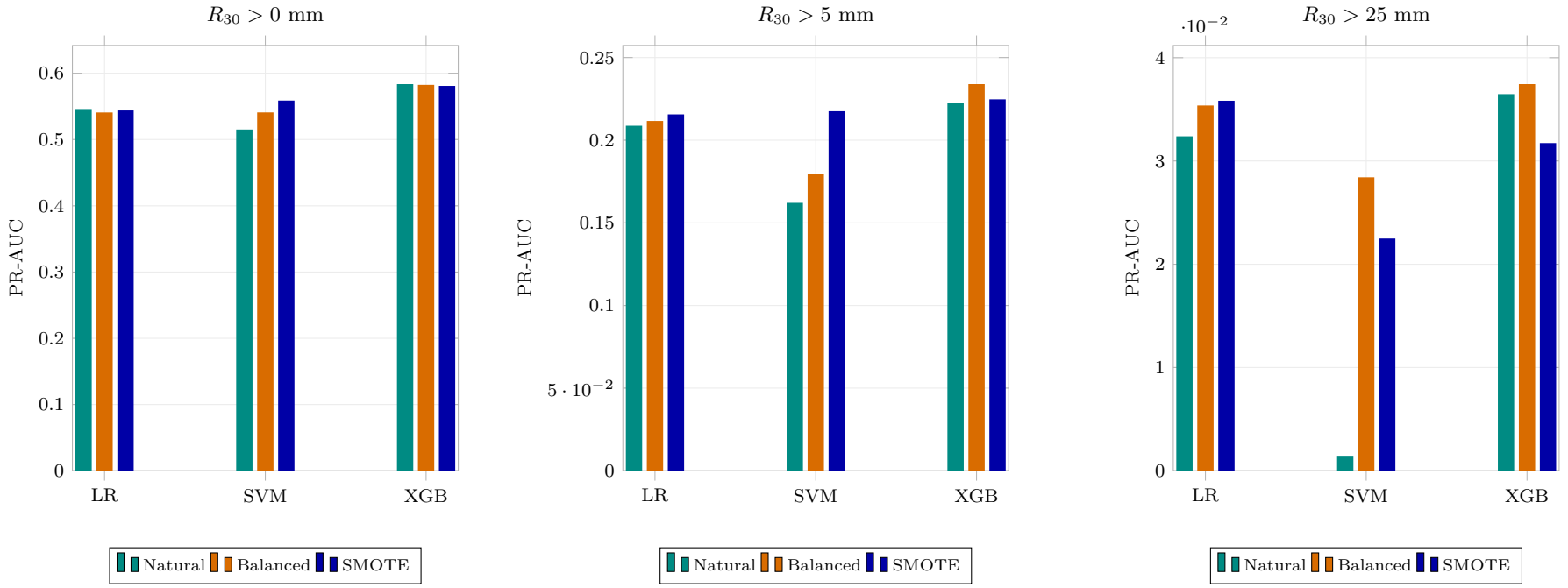
Model selection. The original comparison selects algorithms and policies by 2023 H2 PR-AUC, with Brier as tie-breaker. Final grid hyperparameters use 2022 PR-AUC, while calibrated versus uncalibrated pipelines use 2023 H2 Brier. All decision cutoffs maximize F1 on the corresponding validation predictions and determine test Precision, Recall, F1 and Accuracy; they differ from rainfall thresholds defining labels. The 2024 results do not enter this round of selection.

Algorithm / policy	Selection PR-AUC: 0 mm	5 mm	25 mm
Logistic / Natural	0.54605	0.20874	0.03239
Logistic / Balanced	0.54088	0.21162	0.03537
Logistic / SMOTE	0.54389	0.21562	0.03583
SVM / Natural	0.51511	0.16211	0.00145
SVM / Balanced	0.54095	0.17952	0.02842
SVM / SMOTE	0.55882	0.21753	0.02249
XGBoost / Natural	0.58363	0.22270	0.03647
XGBoost / Balanced	0.58252	0.23392	0.03744
XGBoost / SMOTE	0.58096	0.22471	0.03173

Iteration status. Among the nine main logistic models, 0 reach the 100-iteration cap; 9 of the nine SVM models reach that cap.

Selected models. The original algorithm comparison selects XGBoost for all targets: natural proportions for 0 mm and class weights for 5 and 25 mm. Figures 7 and 8 retain this original comparison. The final prediction interface uniformly uses the three grid-search models in Section 7.1, with their metrics and curves presented separately below.

Figure 7: training policies on validation



Validation comparison. Relative to natural-distribution training, weighting improves PR-AUC in 7 of 9 algorithm/target combinations and SMOTE in 6. Evaluate imbalance treatment by algorithm and target. Every comparison uses the same complete training period and identical validation observations.

7.1. XGBoost grid search and final models

Grid design. Fixing the XGBoost algorithm, each rainfall target traverses the full Cartesian product of four parameter sets: $2^4 = 16$ combinations per target and 48 candidate fits overall. The numbers of trees are 100 or 200, depths 4 or 6, learning rates 0.05 or 0.1, and L2 penalties 1 or 5. Fixed settings are 128 bins, minimum child weight 20, L1 penalty 0, and row/column sampling fractions 1. The 0 mm target uses natural proportions; 5 and 25 mm use class weights. Validation is chronological holdout, not random-fold cross-validation. No training rows are subsampled and no early stopping is used.

Parameter selection and refitting. Each candidate uses all eligible 2017–2021 rows and is ranked by exact Spark PR-AUC on 2022, with Brier as tie-breaker. Preprocessing and classifiers are then refitted on all eligible 2017–2022 rows. Monotone Platt calibration fits 2023 H1; 2023 H2 selects calibrated versus uncalibrated pipelines by Brier and decision cutoffs by F1. Final models are frozen before reading this round of 2024 test results.

Rainfall threshold mm	Trees	Depth	Learning rate	L2 penalty	Final probabilities	2022 PR-AUC
0	100	6	0.05	5	Uncalibrated	0.61137
5	100	6	0.05	5	Platt calibrated	0.25174
25	100	4	0.05	5	Platt calibrated	0.03050

8. Test results in 2024

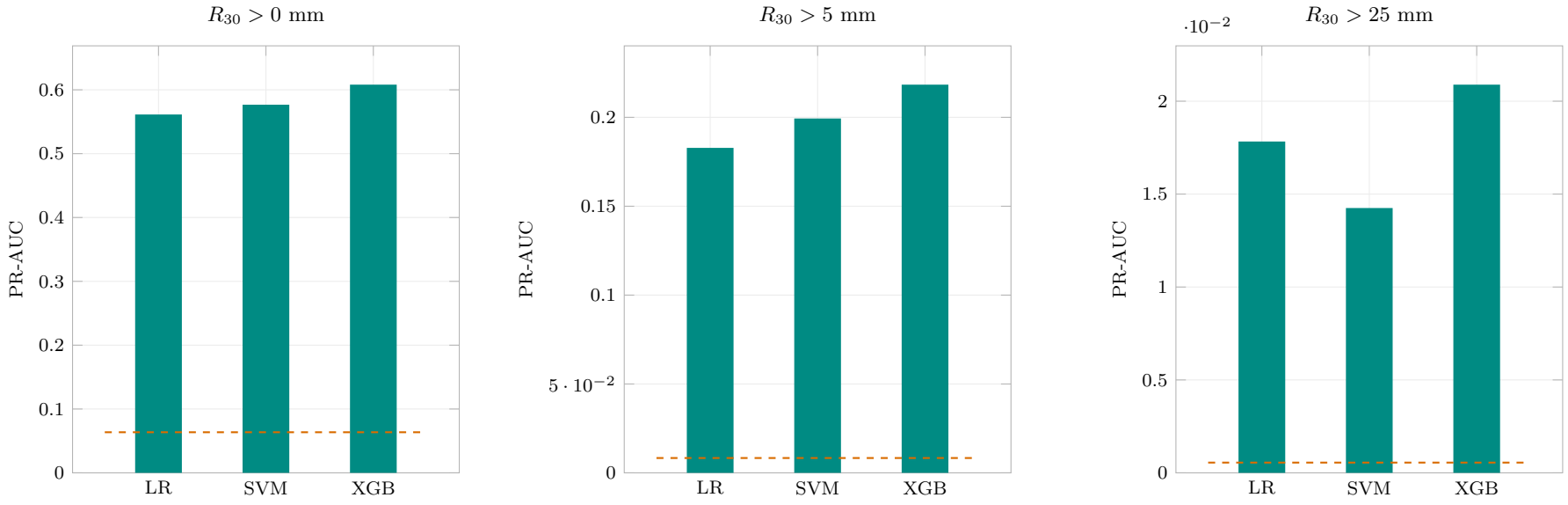
Threshold mm	Original algorithm	Validation-selected policy	ROC-AUC	PR-AUC	Brier
0	Logistic	Natural	0.85625	0.56134	0.0392597
0	SVM	SMOTE	0.85451	0.57665	0.0401863
0	XGBoost	Natural	0.87908	0.60809	0.0364909
5	Logistic	SMOTE	0.85293	0.18282	0.0075397
5	SVM	SMOTE	0.72667	0.19935	0.0077310
5	XGBoost	Balanced	0.87952	0.21839	0.0072009
25	Logistic	SMOTE	0.86018	0.01783	0.0005496
25	SVM	Balanced	0.85936	0.01425	0.0005487
25	XGBoost	Balanced	0.90435	0.02089	0.0005449

Test population. The upper table reports the original algorithm comparison; the lower table reports final grid models. Each target evaluates 6,374,476 eligible 2024 windows, with positive proportions 6.3602%, 0.83447% and 0.055173%. Final 0 mm probabilities are uncalibrated; 5 and 25 mm are calibrated.

Threshold mm	Grid model	Dataset	ROC-AUC	PR-AUC	Brier
0	XGBoost	2023 H2 validation	0.88141	0.58987	0.0343148
0	XGBoost	2024 test	0.88002	0.61218	0.0361268
5	XGBoost	2023 H2 validation	0.89386	0.23927	0.0070153
5	XGBoost	2024 test	0.88159	0.22511	0.0071700
25	XGBoost	2023 H2 validation	0.91441	0.03934	0.0007187
25	XGBoost	2024 test	0.90309	0.02118	0.0005448

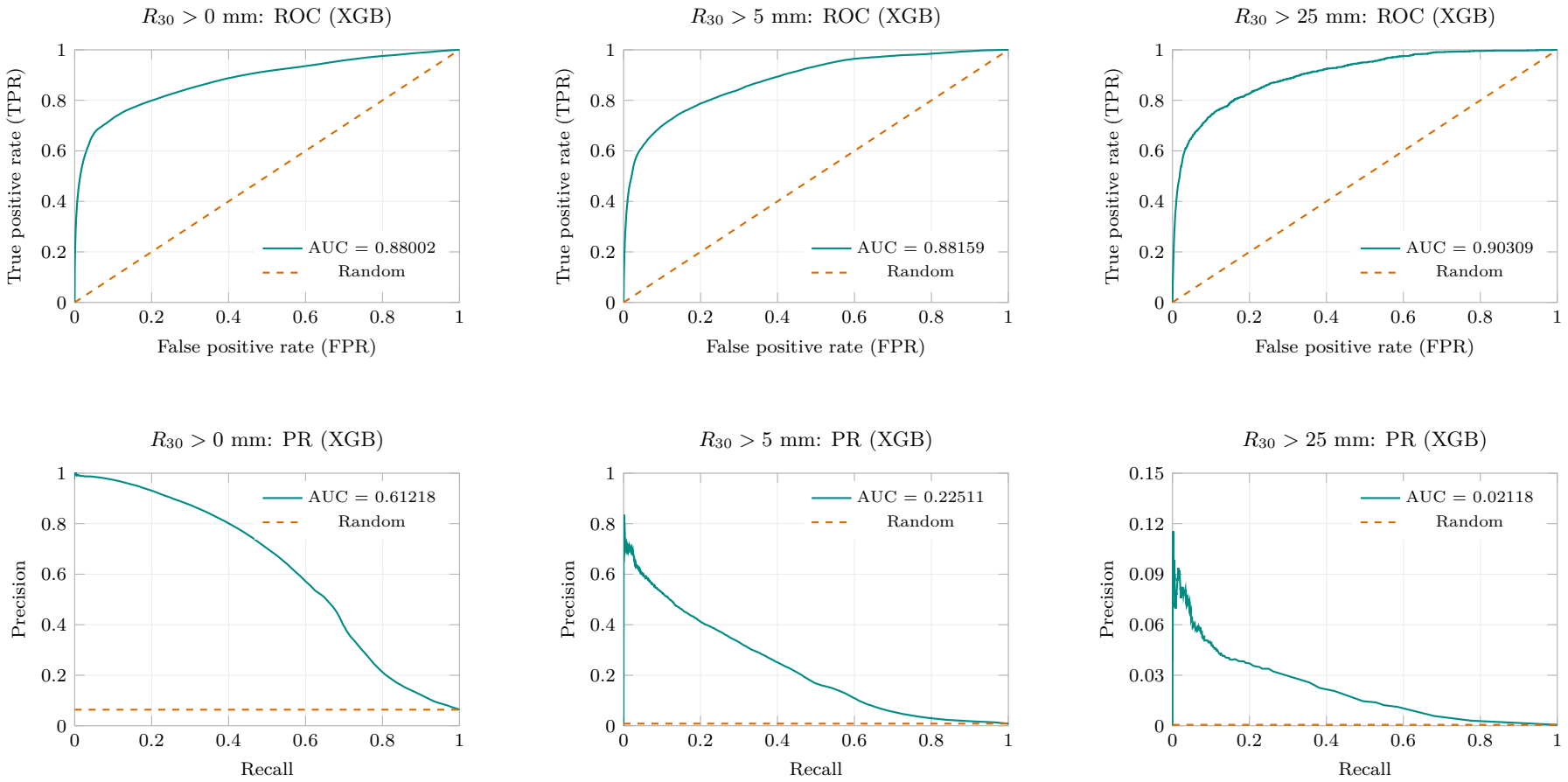
Across-period comparison. The 2022 metrics belong to candidates trained on 2017–2021; 2023 H2 and 2024 metrics belong to frozen models refitted on 2017–2022. Positive proportions differ between periods, so PR-AUC changes reflect both models and event frequency, rather than directly proving improvement or deterioration.

Figure 8: test PR-AUC in the original algorithm comparison



Test comparison. Figure 8 shows the original algorithm comparison, where XGBoost test PR-AUCs are 0.60809, 0.21839 and 0.02089. Final grid values are 0.61218, 0.22511 and 0.02118, all slightly higher; however, 25 mm ROC-AUC and F1 decline, so this is not an improvement across all measures. Final 5 mm PR-AUC also falls below the earlier individually tuned model at 0.22640. Test results are reported without reselecting parameters or models.

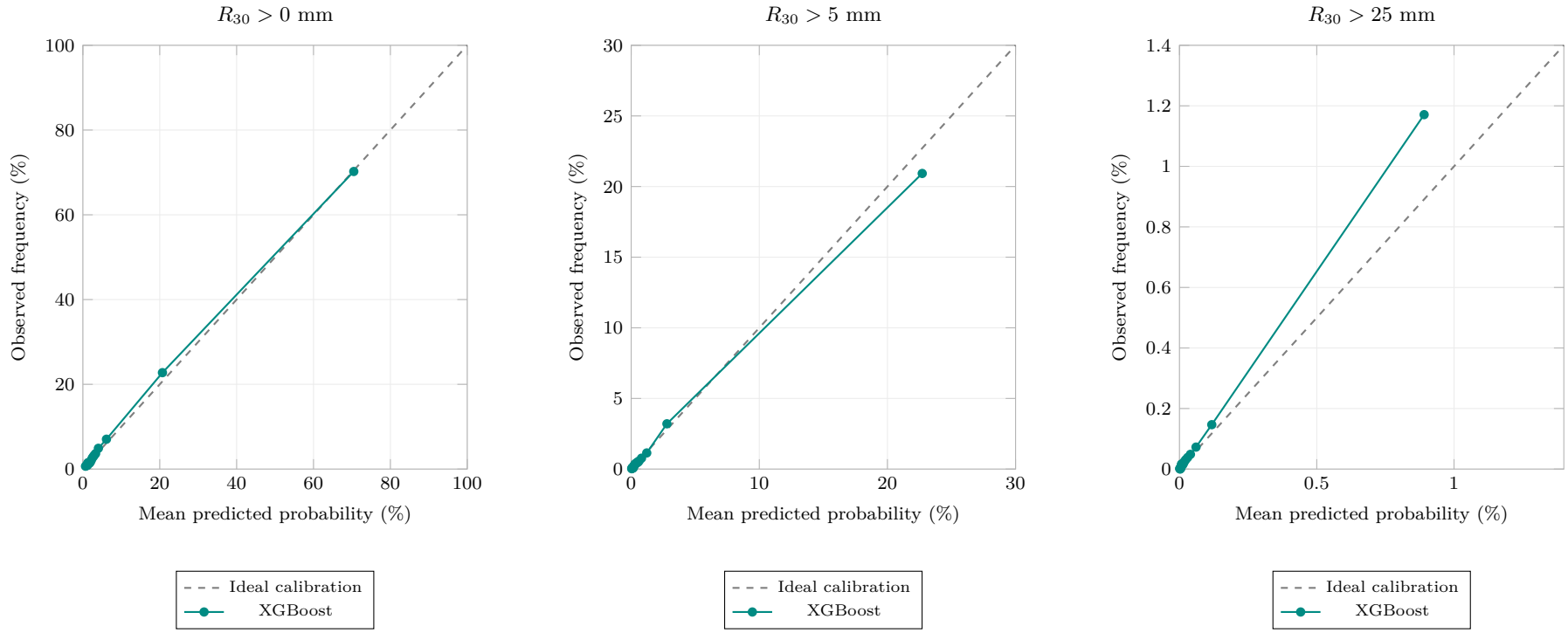
Figure 9: ROC and PR curves of the final selected models



Evaluation objects. Curves correspond to frozen grid models for 0, 5 and 25 mm, refitted on 2017–2022 and evaluated on every eligible 2024 window. Plotted coordinates retain actual curve points through bounded simplification; AUCs use exact full-score integration. ROC reference lines are $y = x$; PR references are test prevalence. The 25 mm PR axis is enlarged to 0–0.15.

Ranking performance. Final ROC-AUCs are 0.88002, 0.88159, 0.90309, and PR-AUCs are 0.61218, 0.22511, 0.02118. Higher rainfall thresholds correspond to rarer events; high ROC-AUC must be interpreted alongside PR curves and prevalence.

Figure 10: reliability and probability quality



Probability quality. Mean predicted probabilities are 6.1816%, 0.8603%, 0.0426%, versus observed proportions 6.3602%, 0.8345% and 0.0552%. Each point aggregates windows with similar probabilities; grey lines represent ideal calibration. Similar overall means do not guarantee accuracy at every risk level.

Calibration selection. On 2023 H2, 0 mm uncalibrated Brier is 0.0343148, better than calibrated 0.0344561, so uncalibrated probabilities are retained. Calibrated Brier for 5 and 25 mm is 0.0070153 and 0.0007187, substantially below uncalibrated weighted-model errors of 0.1144999 and 0.0994519. Pipeline selection uses validation only; testing evaluates the chosen results.

Rare-event calibration. The highest mean-probability group for 25 mm contains 170,814 windows, with mean prediction 0.891% and observed frequency 1.171%. Interpret this group alongside limited positive counts. No event-block confidence intervals are computed, so no statistically significant improvement is claimed.

9. Classification metrics and further analysis

Threshold mm	Grid model	Decision cutoff	Precision	Recall	F1	Accuracy
0	XGBoost	0.285102	0.62658	0.56163	0.59233	0.95083
5	XGBoost	0.194984	0.30804	0.32470	0.31615	0.98828
25	XGBoost	0.031623	0.04585	0.10918	0.06458	0.99825

Classification evaluation. Cutoffs are validation-selected. Negative cases dominate high-threshold targets, so high accuracy alone does not establish effective heavy-rain detection. Compare ROC-AUC and PR-AUC primarily, with Brier and reliability for probabilities.

Original SMOTE counts. The following table belongs to the original comparison; final grid models use no synthetic observations.

Threshold mm	Real positives	Added rows	Augmented positives (%)	Changed rain rows	Isolated copies
0	2,074,430	6,042,534	20.000	3,615,180	263,540
5	279,801	2,518,209	7.550	1,292,705	291,565
25	19,787	178,083	0.570	67,189	54,652

Spatial and temporal dependence. Neighbouring stations can share a rainfall event and half-hour windows overlap. Event/day-grouped evaluation can assess stability across stations and events. Independent threshold classifiers also motivate future monotone cross-threshold probability constraints.

Earlier tuning and spatial comparison. The earlier six-configuration 5 mm search achieved 2023 H2 PR-AUC 0.239991 and Brier 0.00701053, slightly better than final grid-model values 0.239268 and 0.00701530. The project nevertheless fixes the grid model for a uniform three-target procedure and does not claim it outperforms the earlier 5 mm model. Adding eight spatial interaction features at the earlier fixed parameters produced 2023 H2 PR-AUC 0.238817 and Brier 0.00701540, so that extension was not adopted.

Prediction interface and limitations. Station, prediction time on the five-minute grid and a threshold of 0, 5 or 25 mm select the corresponding grid pipeline and exceedance probability; the original comparison model remains explicitly selectable. Prepared historical feature rows are still required, and their sample scope depends on complete future labels; this is not yet a live service. Targets are trained independently without cross-threshold probability monotonicity guarantees. The original experiment previously inspected 2024; this round freezes parameters and models before retesting, but remains a follow-up experiment rather than a researcher-unseen sealed test.

9.1. Supplementary experiment: distance-aware five-neighbour features

Purpose and boundary. This is a supplementary experiment alongside the reviewed grid-XGBoost main line. The default prediction interface and primary conclusions remain the grid models in Section 7.1. It asks whether neighbouring gauges have different predictive value when their distances differ. Every feature uses only station coordinates and five-minute readings available at the prediction time; no future rainfall, validation rows or test rows enter feature construction.

Features and training. The original 73-dimensional tree vector receives 14 measurements/availability variables for five nearest neighbours and their 14 missingness indicators (28 dimensions), then five neighbour distances, nearest reporting-neighbour distance, inverse-distance-weighted rainfall, exponentially decayed rainfall with a 5 km scale, and their eight missingness indicators (16 dimensions), giving 117 dimensions. Distances and graph edges use station coordinates known at the time; imputation values fit on training rows only. The experiment trains on 34,542,285 eligible 2017–2022 windows, calibrates on 2023 H1, selects among 27 algorithm/class-policy candidates on 2023 H2, and evaluates frozen models on all of 2024. Full distance training ran on a Google Cloud Compute Engine Ubuntu 24.04 `e2-custom-20-65536` VM (20 vCPU, 64 GiB RAM and 200 GB persistent disk), using Python 3.11.10, Java 17.0.20.1, Spark 3.5.8 and XGBoost 3.2.0.

Frozen-model results for 2024. XGBoost is selected for all three thresholds: natural class proportions at 0 mm and class weighting at 5 and 25 mm.

Threshold (mm)	Grid PR-AUC	Grid ROC-AUC	Grid Brier	Distance PR-AUC	Distance ROC-AUC	Distance Brier
0	0.61218	0.88002	0.036127	0.62838	0.89045	0.035469
5	0.22511	0.88159	0.007170	0.23991	0.89277	0.007079
25	0.02118	0.90309	0.000545	0.02367	0.91855	0.000544

Interpretation. In this supplementary experiment, the distance version has higher PR-AUC and ROC-AUC and lower Brier score at every threshold in 2024, so explicit distance information merits further study. The model families differ in features, parameter search and model-selection protocol; this table is a supplementary descriptive comparison, not evidence attributable to one feature or a test-set-driven replacement of the default model. The 2024 period had already been inspected in the original experiment, so this is follow-up evaluation after the specification was frozen, rather than a fully unseen sealed test.

10. AI declaration and references

AI-use declaration. We use OpenAI Codex to assist study design, data-processing and model code, result checks and report editing. Reported rainfall statistics and model evaluations come from actual processing and model runs; SMOTE examples are used only for training.

Sources.

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